BARBADOS NATIONAL REGISTRY

Annual cardiovascular disease report

2025

Cardiovascular disease events and mortality in Barbados

The University of the West Indies | Cave Hill Campus

About this report

Annual cardiovascular disease report 2025

Cardiovascular disease events and mortality in Barbados

Purpose and scope

This report provides an annual overview of cardiovascular disease events and mortality in Barbados. It presents published All-CVD, Heart and Stroke results, together with selected sex and age summaries where those measures are available in the approved public release. The report is intended to support decision-making by government, hospitals, clinics, public-health teams, researchers and other partners.

The report uses complete annual observations from the approved public event and mortality releases. It does not read confidential source records or recalculate the published surveillance measures. Estimates should be interpreted alongside the definitions, coverage notes and uncertainty information provided in the report.

Report information

Report	Annual cardiovascular disease report 2025
Prepared by	Christina Howitt; Ian Hambleton; Jacqueline Campbell
Institution	Barbados National Registry | The University of the West Indies, Cave Hill Campus
Approved data releases	CVD events: cvd_2026_01 | CVD mortality: mort_2026_07
Online report	https://uwi-bnr.github.io/info-hub/
Suggested citation	Barbados National Registry. Annual cardiovascular disease report 2025: cardiovascular disease events and mortality in Barbados. The University of the West Indies, Cave Hill Campus.

Acknowledgements

The Barbados National Registry acknowledges the clinical, hospital, registry, data-management and public-health teams whose work supports the production of this surveillance report. The report should be read as a surveillance product and not as a substitute for clinical judgement, formal cause-of-death coding or other official statistical series.

Further information

Definitions, ascertainment rules, mortality classifications, uncertainty measures, data availability and disclosure controls are described in the Methods chapter. The latest public report and associated data products are available at: https://uwi-bnr.github.io/info-hub/

Publication note: This edition is produced through the controlled BNR annual-report build, approval and publication workflow. The public version is the approved version deposited through that workflow.

2025 in brief

The latest complete year at a glance

CVD EVENTS

1,116

Primary national estimate

CVD EVENT RATE

229 per 100,000

Age-standardised | 95% CI 215-244

CVD DEATHS

368

Primary definition

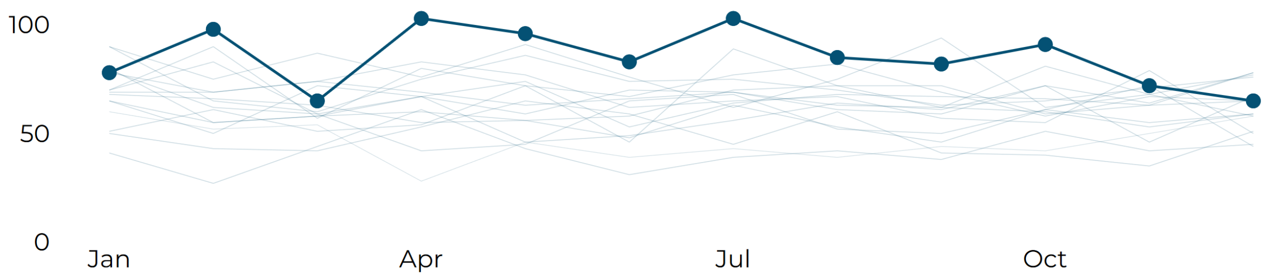
CVD MORTALITY RATE

76 per 100,000

Age-standardised | 95% CI 68-85

How the year unfolded

Monthly hospital-recorded All-CVD events in 2025. National estimates are annual because death-record ascertainment is applied after the year closes.



What stood out in 2025?

01 There were 1,021 hospital-recorded CVD events in 2025, about 27% above the published previous-five-year mean of 805.8. The Primary national estimate was 1,115.8 events after adding the estimated contribution from events identified through death records.

02 Stroke accounted for 67.9% of hospital-recorded CVD events in 2025. The Primary national age-standardised event rate was also higher in men than women: 288.8 compared with 184.0 per 100,000.

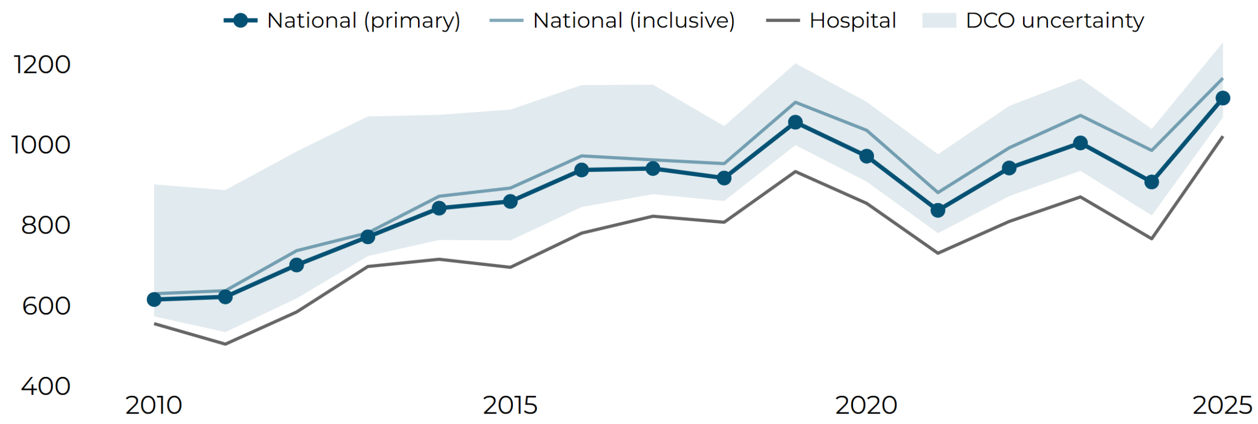
03 There were 368 Primary CVD deaths in 2025, about 7% below the previous-five-year mean of 394.2. The Inclusive definition counted 588 deaths, close to its recent five-year comparator, showing how strongly mortality totals depend on whether Possible deaths are included.

1 | CVD events

CVD event counts

Hospital-recorded events and published national estimates across the complete annual series.

Hospital-recorded	Primary national	Inclusive national
1,021	1,116	1,165



Measure	2021	2022	2023	2024	2025
Hospital-recorded	730	809	870	766	1,021
Primary national	837	942	1,004	907	1,116
Inclusive national	881	992	1,073	986	1,165

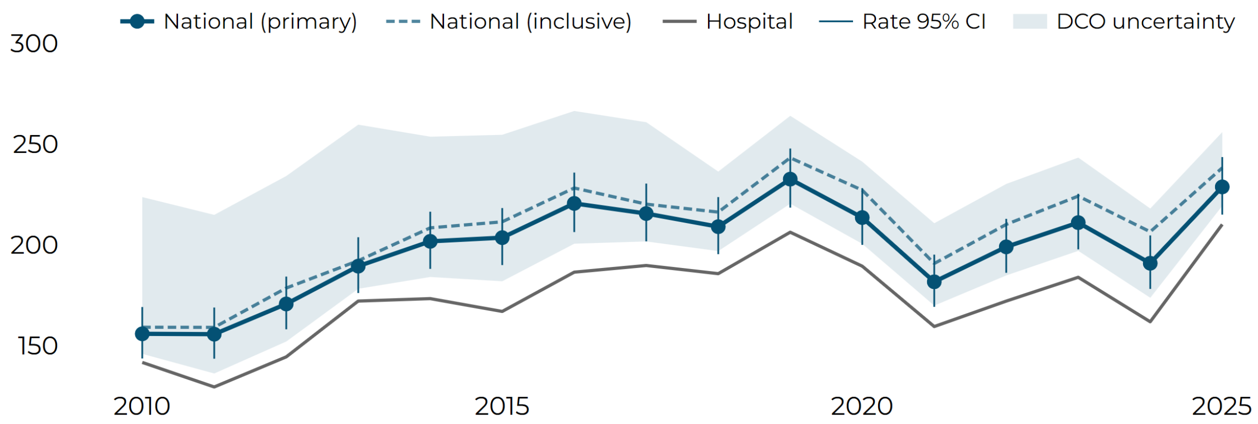
WHAT THIS MEANS

Hospital-recorded CVD events were higher in 2025 than in the recent comparison period: 1,021 events compared with a published previous-five-year mean of 805.8. The Primary national estimate was 1,115.8 and the Inclusive estimate 1,165.4, so adding death-record ascertainment increases the estimated national count without changing the broad message that 2025 was a high-count year relative to the recent hospital series.

CVD event rates

Age-standardised rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Hospital-recorded ASR	Primary national ASR	Inclusive national ASR
210.1	**228.8**	**238.3**



Latest five complete years

ASR per 100,000	2021	2022	2023	2024	2025
Hospital-recorded estimate	159.4	172.1	183.9	161.8	210.1
Hospital-recorded 95% CI	147.7 - 172.2	160.1 - 185.1	171.5 - 197.3	150.1 - 174.6	196.9 - 224.4
Primary national estimate	181.7	199.0	211.0	190.9	228.8
Primary national 95% CI	169.2 - 195.1	186.1 - 212.9	197.8 - 225.3	178.1 - 204.7	215.0 - 243.6
Inclusive national estimate	190.7	210.2	224.2	206.5	238.3
Inclusive national 95% CI	178.0 - 204.5	196.9 - 224.5	210.6 - 238.9	193.3 - 220.7	224.2 - 253.3

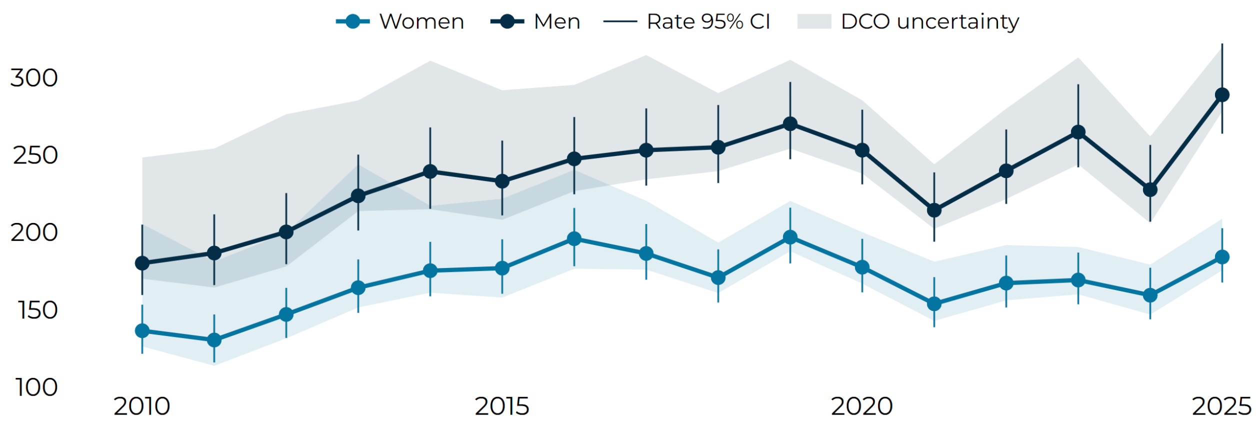
WHAT THIS MEANS

The 2025 Primary national age-standardised CVD event rate was 228.8 per 100,000, compared with 210.1 for the hospital-recorded series and 238.3 for the Inclusive national estimate. Statistical confidence intervals and DCO linkage ranges overlap substantially across these definitions, so the report treats the differences as uncertainty around ascertainment rather than as separate disease trends.

CVD events: women and men

Primary national age-standardised event rates per 100,000.

All sexes	Women	Men
228.8	184.0	288.8



Latest five complete years

Primary ASR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	181.7	199.0	211.0	190.9	228.8
All sexes 95% CI	169.2 - 195.1	186.1 - 212.9	197.8 - 225.3	178.1 - 204.7	215.0 - 243.6
Women estimate	153.7	167.1	169.1	159.3	184.0
Women 95% CI	138.6 - 170.9	151.3 - 184.9	153.4 - 186.8	143.7 - 177.0	167.5 - 202.6
Men estimate	214.2	239.7	264.7	227.5	288.8
Men 95% CI	193.9 - 238.6	218.3 - 266.4	241.9 - 295.6	206.8 - 256.4	263.7 - 322.0

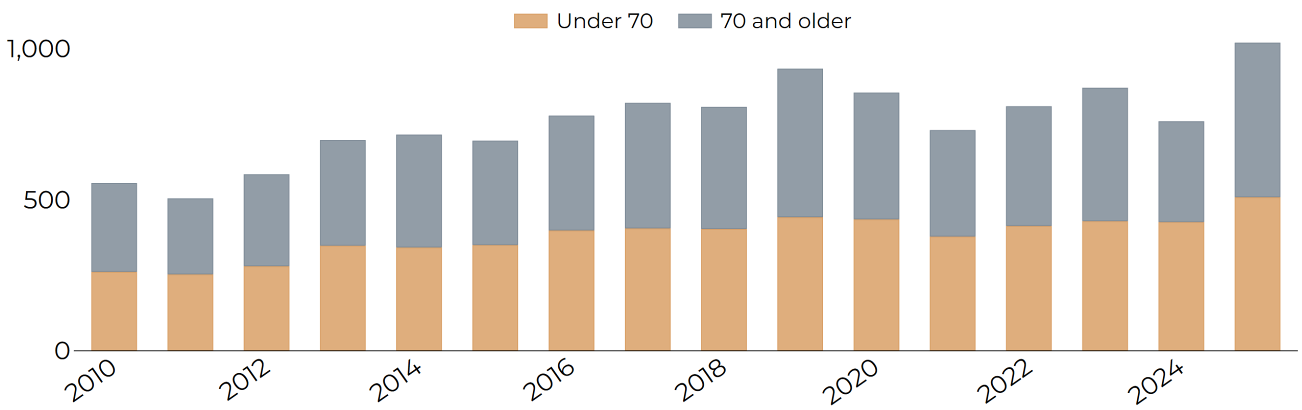
WHAT THIS MEANS

Men had a higher Primary national age-standardised CVD event rate than women in 2025: 288.8 compared with 184.0 per 100,000. The difference is also visible in the hospital-recorded series, so it is not explained solely by the addition of death-record-only events.

CVD events: age patterns

Hospital-recorded annual event counts by broad age group. These are composition counts, not age-specific rates.

All ages	Under 70	70 and older
1,021	508	511



Latest five complete years

Hospital-recorded events	2021	2022	2023	2024	2025
All ages	730	809	870	766	1,021
Under 70	378	413	429	426	508
70 and older	352	396	441	333	511

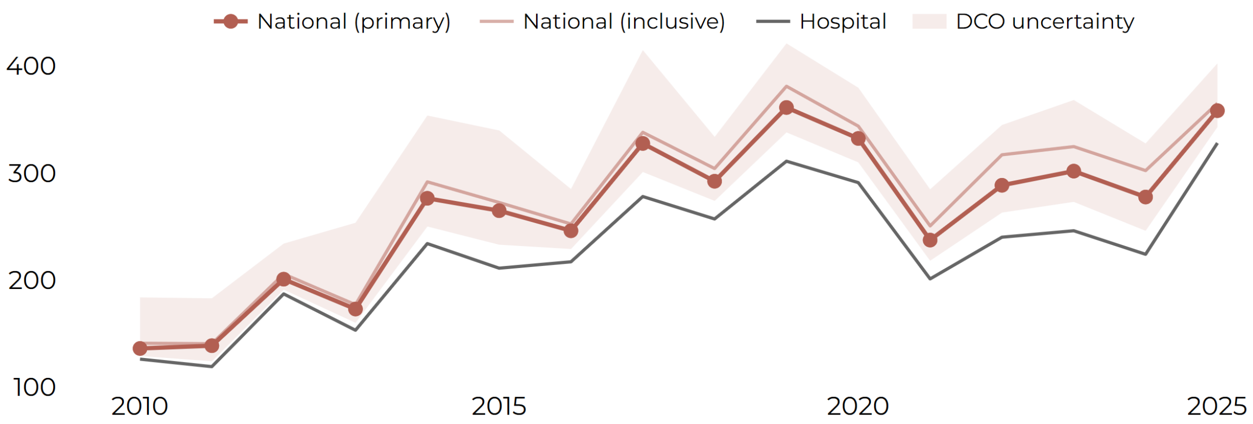
WHAT THIS MEANS

The 2025 hospital-recorded event count was almost evenly divided by age: 508 events were among people aged under 70 and 511 among people aged 70 or older. This age profile describes the composition of recorded events; it should not be read as an age-specific population risk comparison.

Heart event counts

Hospital-recorded events and published national estimates across the complete annual series.

Hospital-recorded	Primary national	Inclusive national
328	358	365



Measure	2021	2022	2023	2024	2025
Hospital-recorded	201	240	246	224	328
Primary national	237	289	302	278	358
Inclusive national	251	317	325	302	365

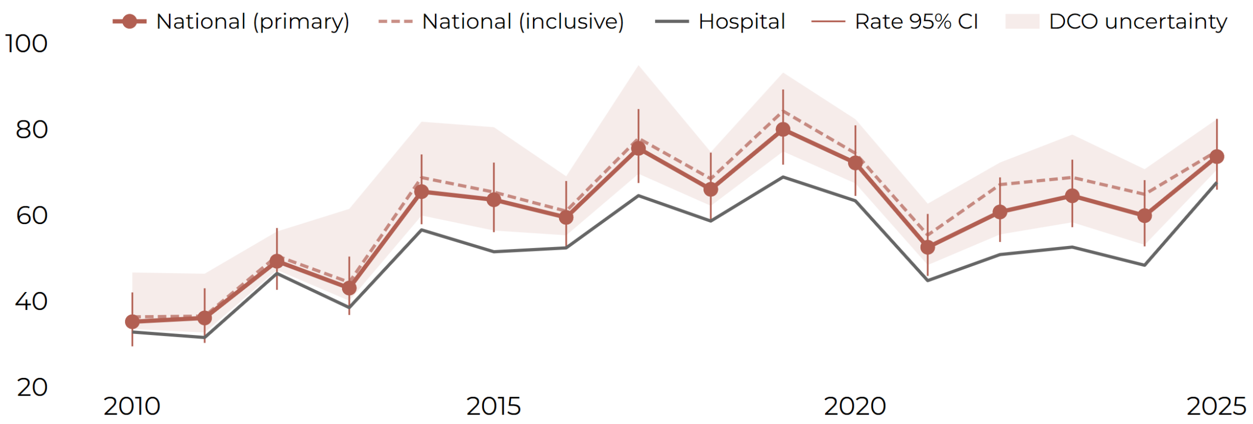
WHAT THIS MEANS

Heart events form the smaller of the two main event groups shown on the preceding CVD pages. The chart and table show how the hospital-recorded series compares with the published national estimates; the definitions should be compared within this page, rather than added to the Stroke rows.

Heart event rates

Age-standardised rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Hospital-recorded ASR	Primary national ASR	Inclusive national ASR
67.7	**73.7**	**74.9**



Latest five complete years

ASR per 100,000	2021	2022	2023	2024	2025
Hospital-recorded estimate	44.8	50.8	52.6	48.3	67.7
Hospital-recorded 95% CI	38.6 - 52.0	44.5 - 58.3	46.0 - 60.2	41.9 - 55.9	60.3 - 76.2
Primary national estimate	52.5	60.8	64.5	59.9	73.7
Primary national 95% CI	45.9 - 60.3	53.8 - 68.8	57.2 - 72.9	52.7 - 68.2	65.9 - 82.4
Inclusive national estimate	55.4	67.1	68.8	64.9	74.9
Inclusive national 95% CI	48.5 - 63.3	59.7 - 75.6	61.3 - 77.4	57.4 - 73.4	67.2 - 83.8

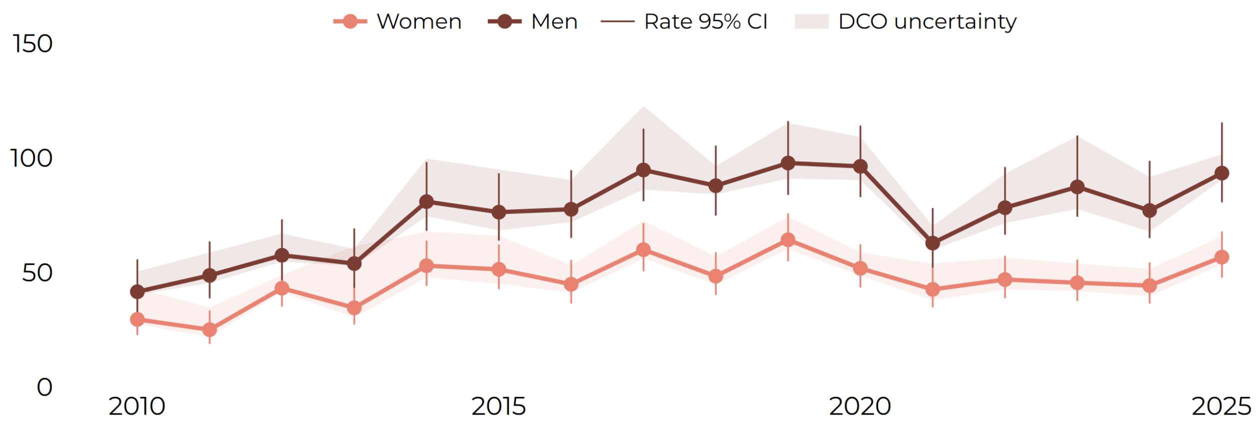
WHAT THIS MEANS

This page shows the recent pattern in Heart event rates after allowing for differences in the age structure of the population. Read the solid estimate and its confidence interval together; small year-to-year movements may not represent a meaningful change.

Heart events: women and men

Primary national age-standardised event rates per 100,000.

All sexes	Women	Men
73.7	56.7	93.4



Latest five complete years

Primary ASR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	52.5	60.8	64.5	59.9	73.7
All sexes 95% CI	45.9 - 60.3	53.8 - 68.8	57.2 - 72.9	52.7 - 68.2	65.9 - 82.4
Women estimate	42.6	46.9	45.5	44.3	56.7
Women 95% CI	34.8 - 52.7	38.7 - 57.3	37.5 - 55.7	36.4 - 54.4	47.7 - 67.9
Men estimate	62.8	78.3	87.4	77.1	93.4
Men 95% CI	52.1 - 78.1	66.5 - 96.1	74.3 - 109.8	65.0 - 98.7	80.6 - 115.5

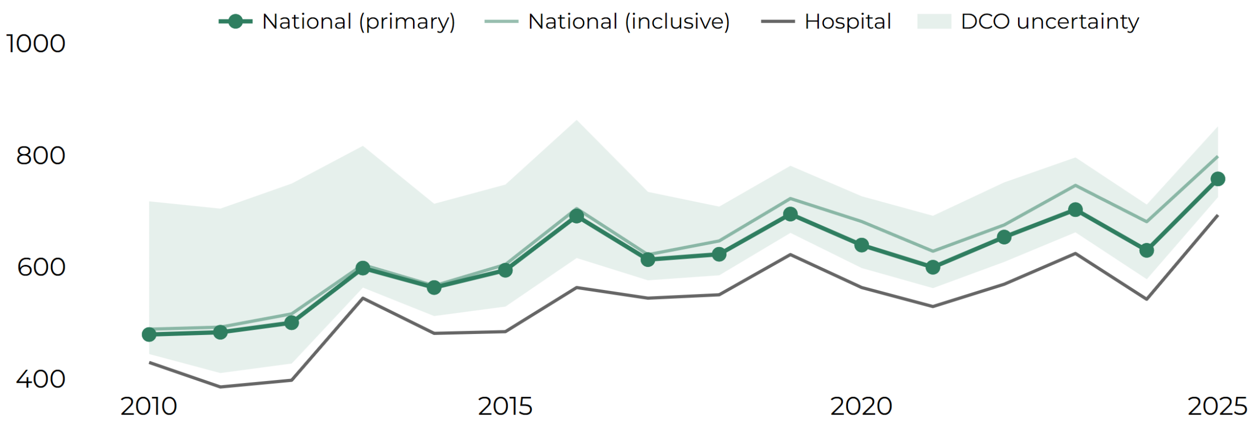
WHAT THIS MEANS

The comparison helps services ask whether Heart events are affecting women and men differently after age is taken into account. It is a signal for planning and prevention, not evidence on its own of why a difference has occurred.

Stroke event counts

Hospital-recorded events and published national estimates across the complete annual series.

Hospital-recorded	Primary national	Inclusive national
693	758	798



Measure	2021	2022	2023	2024	2025
Hospital-recorded	529	569	624	542	693
Primary national	599	653	702	630	758
Inclusive national	628	675	746	681	798

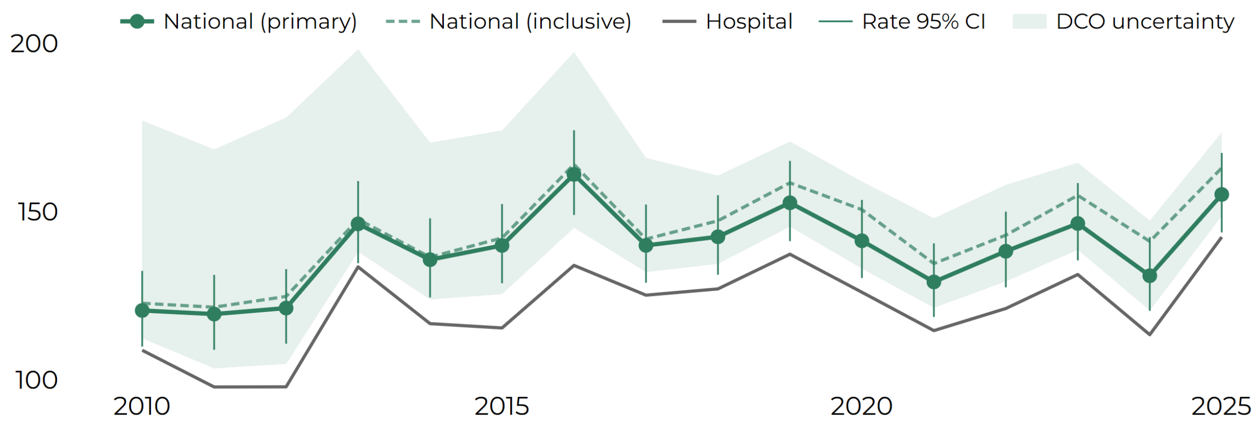
WHAT THIS MEANS

Stroke remains the larger of the two main event groups shown on the preceding CVD pages. The national estimates add published death-record ascertainment to the hospital series, helping services see the likely scale of events beyond hospital records alone.

Stroke event rates

Age-standardised rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Hospital-recorded ASR	Primary national ASR	Inclusive national ASR
142.5	**155.1**	**163.0**



Latest five complete years

ASR per 100,000	2021	2022	2023	2024	2025
Hospital-recorded estimate	114.7	121.2	131.3	113.5	142.5
Hospital-recorded 95% CI	104.8 - 125.5	111.2 - 132.3	120.9 - 142.8	103.7 - 124.2	131.6 - 154.4
Primary national estimate	129.1	138.2	146.5	131.0	155.1
Primary national 95% CI	118.7 - 140.6	127.5 - 150.0	135.5 - 158.5	120.5 - 142.4	143.8 - 167.5
Inclusive national estimate	134.6	143.0	154.9	141.2	163.0
Inclusive national 95% CI	124.0 - 146.2	132.1 - 155.0	143.6 - 167.2	130.4 - 153.0	151.4 - 175.6

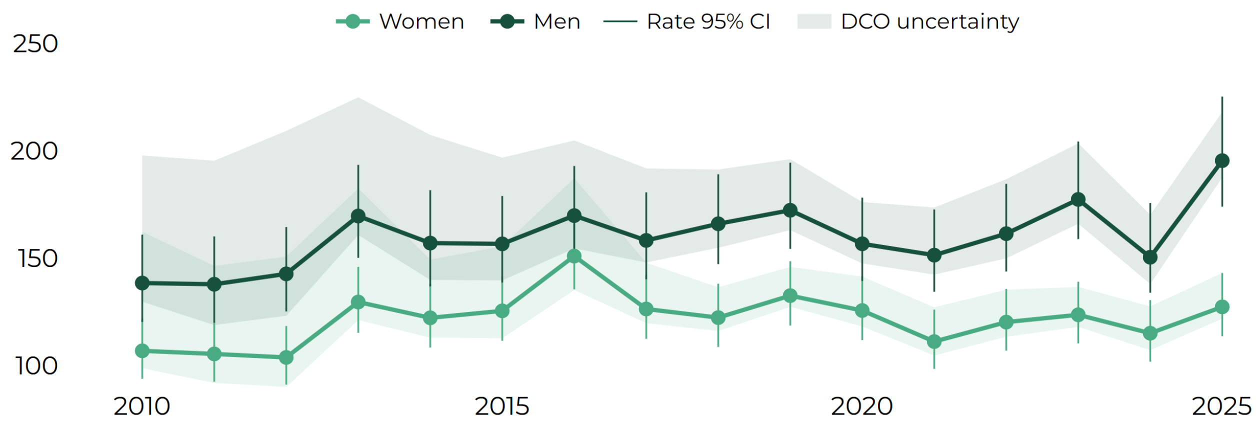
WHAT THIS MEANS

This page shows the recent pattern in Stroke event rates after allowing for differences in the age structure of the population. The wider CVD page provides the overall context; here the focus is the Stroke contribution to that total.

Stroke events: women and men

Primary national age-standardised event rates per 100,000.

All sexes	Women	Men
155.1	127.3	195.4



Latest five complete years

Primary ASR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	129.1	138.2	146.5	131.0	155.1
All sexes 95% CI	118.7 - 140.6	127.5 - 150.0	135.5 - 158.5	120.5 - 142.4	143.8 - 167.5
Women estimate	111.1	120.2	123.6	115.0	127.3
Women 95% CI	98.4 - 126.0	106.9 - 135.6	110.3 - 139.0	101.7 - 130.4	113.6 - 143.1
Men estimate	151.4	161.4	177.4	150.4	195.4
Men 95% CI	134.3 - 172.6	143.7 - 184.6	158.9 - 204.3	133.9 - 175.7	174.0 - 225.2

WHAT THIS MEANS

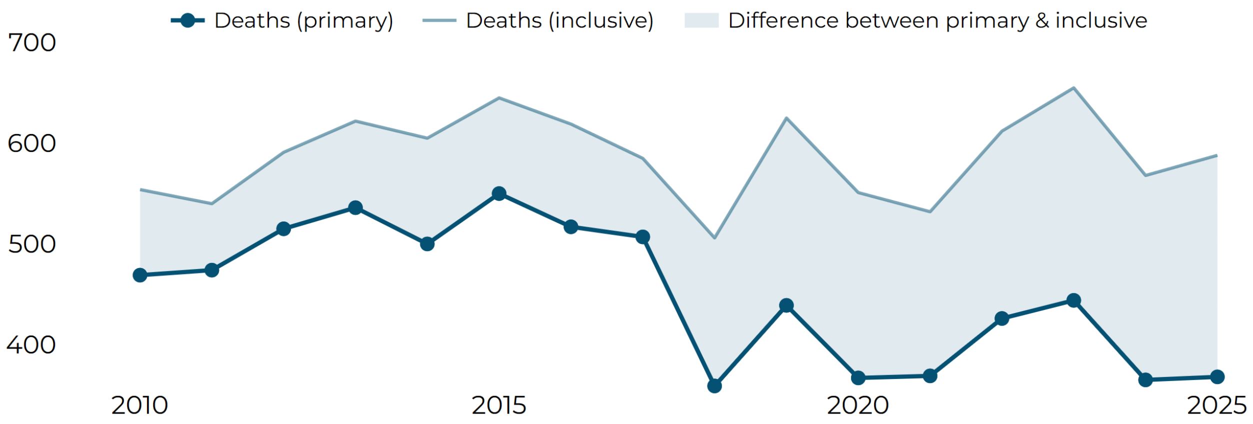
The comparison helps services ask whether Stroke events are affecting women and men differently after age is taken into account. It should be read alongside the all-CVD comparison on the earlier page.

2 | CVD mortality

CVD deaths

Primary = Clear + Likely CVD deaths. Inclusive = Clear + Likely + Possible CVD deaths.

Primary deaths	Inclusive deaths
368	588



Latest five complete years

Deaths	2021	2022	2023	2024	2025
Primary	369	426	444	365	368
Inclusive	532	612	655	568	588

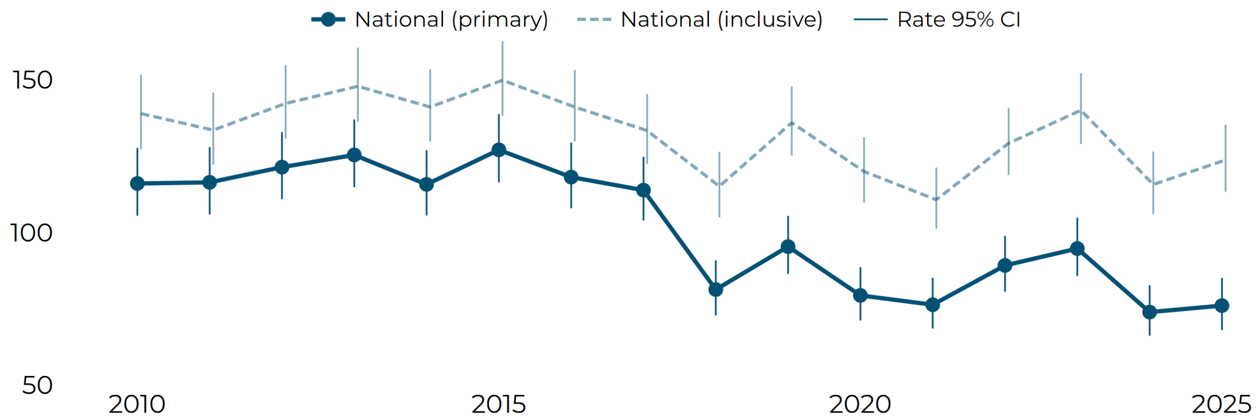
WHAT THIS MEANS

Primary CVD deaths numbered 368 in 2025, below the published previous-five-year mean of 394.2. The Inclusive count was 588, close to its comparator of 583.6. The contrast between those two patterns is important: the recent position depends substantially on whether deaths classified as Possible CVD are included.

CVD mortality rates

Primary and Inclusive age-standardised mortality rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Primary ASMR	Inclusive ASMR
76.0	123.7



Latest five complete years

ASMR per 100,000	2021	2022	2023	2024	2025
Primary estimate	76.3	89.2	94.7	73.9	76.0
Primary 95% CI	68.5 - 85.1	80.5 - 98.9	85.7 - 104.8	66.1 - 82.7	68.0 - 85.1
Inclusive estimate	110.7	129.2	140.0	115.7	123.7
Inclusive 95% CI	101.2 - 121.2	118.7 - 140.8	129.0 - 152.1	105.9 - 126.5	113.3 - 135.2

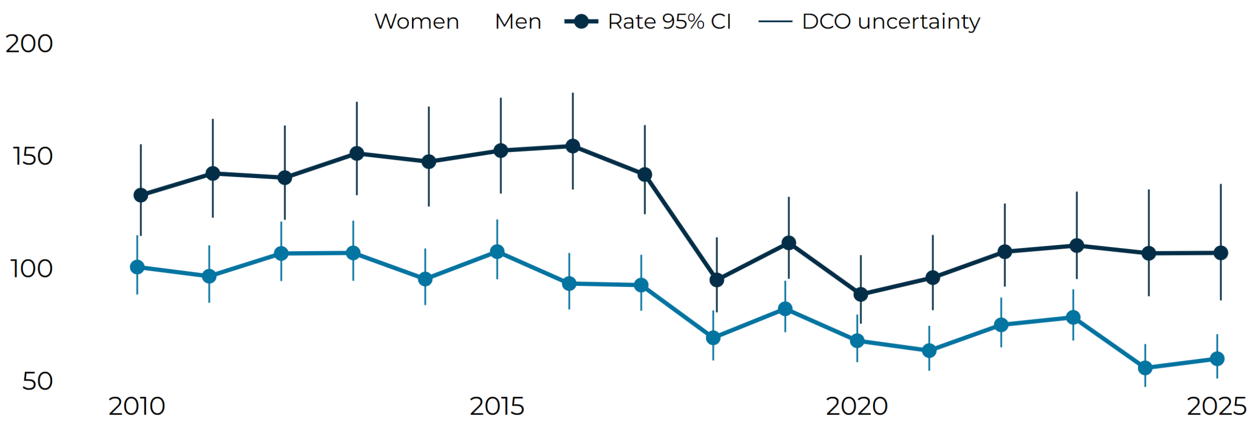
WHAT THIS MEANS

The 2025 Primary age-standardised CVD mortality rate was 76.0 per 100,000, while the Inclusive rate was 123.7. Their statistical confidence intervals are shown separately from the definitional difference; the gap between Primary and Inclusive estimates is not a confidence interval.

CVD deaths: women and men

Primary age-standardised mortality rates per 100,000.

All sexes	Women	Men
76.0	59.7	106.8



Latest five complete years

Primary ASMR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	76.3	89.2	94.7	73.9	76.0
All sexes 95% CI	68.5 - 85.1	80.5 - 98.9	85.7 - 104.8	66.1 - 82.7	68.0 - 85.1
Women estimate	63.4	74.8	78.2	55.7	59.7
Women 95% CI	54.4 - 74.4	64.8 - 87.0	67.8 - 90.6	47.2 - 66.3	50.9 - 70.7
Men estimate	95.8	107.4	110.1	106.7	106.8
Men 95% CI	81.3 - 114.8	91.8 - 128.8	95.2 - 134.1	87.5 - 135.1	85.7 - 137.5

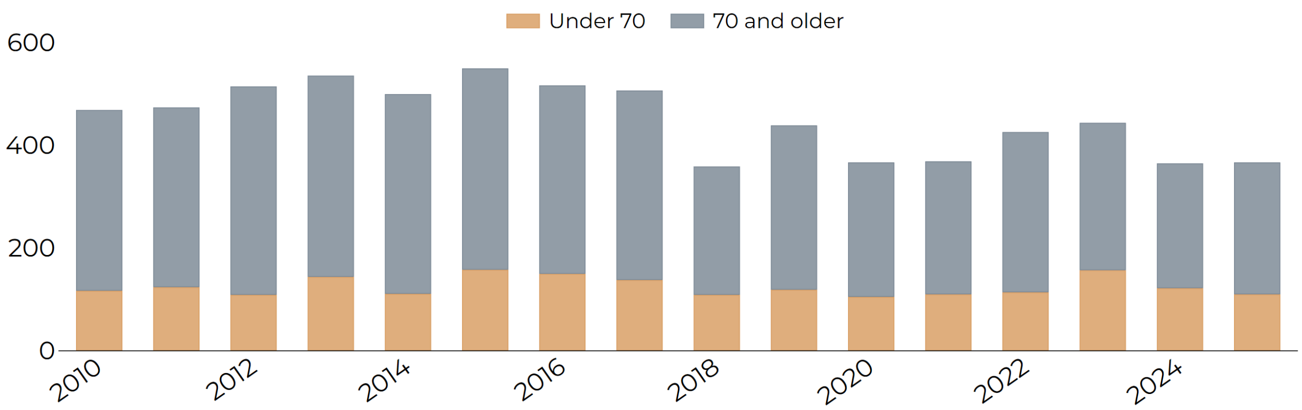
WHAT THIS MEANS

Primary CVD death counts were identical for women and men in 2025 at 184 each, but the age-standardised mortality rate was higher in men: 106.8 compared with 59.7 per 100,000 in women. The published 95% confidence intervals do not overlap, illustrating why rates add information that raw counts alone cannot provide.

CVD deaths: age patterns

Primary-definition annual death counts by broad age group. These are composition counts, not age-specific rates.

All ages	Under 70	70 and older
368	110	257



Latest five complete years

Primary deaths	2021	2022	2023	2024	2025
All ages	369	426	444	365	368
Under 70	110	114	157	122	110
70 and older	259	312	287	243	257

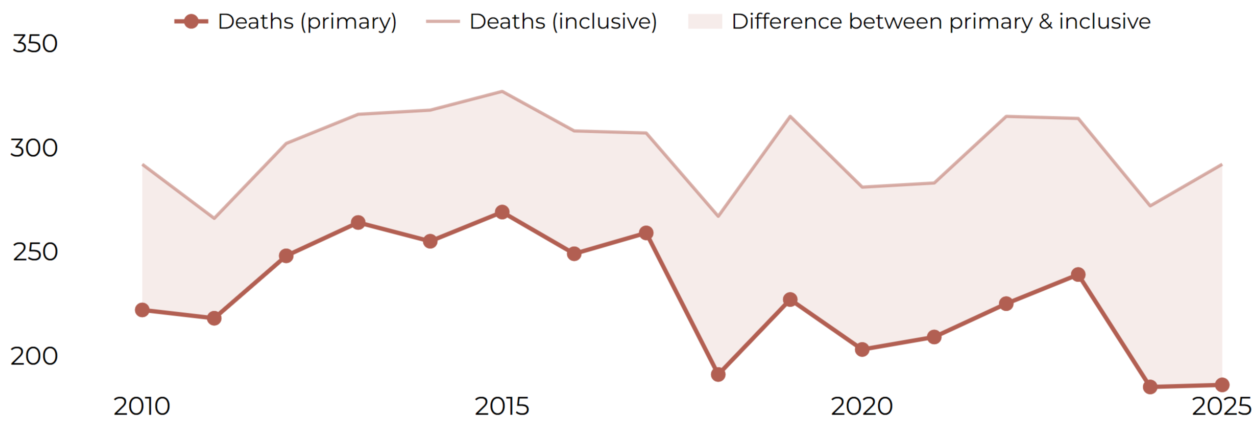
WHAT THIS MEANS

Among Primary CVD deaths with an age classification in 2025, 70.0% were aged 70 or older and 30.0% were under 70. One Primary CVD death is outside that age-distribution denominator, so the age percentages should be read from the published distribution rather than reconstructed from the total count.

Heart deaths

Primary = Clear + Likely CVD deaths. Inclusive = Clear + Likely + Possible CVD deaths.

Primary deaths	Inclusive deaths
186	292



Latest five complete years

Deaths	2021	2022	2023	2024	2025
Primary	209	225	239	185	186
Inclusive	283	315	314	272	292

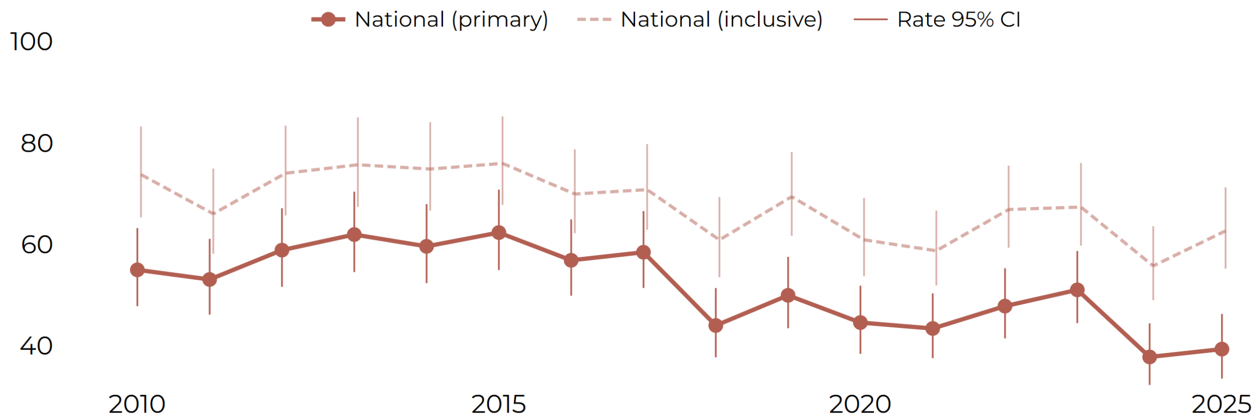
WHAT THIS MEANS

Heart deaths are shown using both the Primary and Inclusive definitions. The difference between them reflects how deaths classified as Possible CVD are handled, rather than a second group of people who died.

Heart mortality rates

Primary and Inclusive age-standardised mortality rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Primary ASMR	Inclusive ASMR
39.3	62.6



Latest five complete years

ASMR per 100,000	2021	2022	2023	2024	2025
Primary estimate	43.4	47.8	51.0	37.8	39.3
Primary 95% CI	37.5 - 50.3	41.4 - 55.3	44.4 - 58.7	32.2 - 44.4	33.5 - 46.2
Inclusive estimate	58.7	66.9	67.3	55.7	62.6
Inclusive 95% CI	51.9 - 66.6	59.3 - 75.5	59.7 - 76.0	49.0 - 63.6	55.2 - 71.2

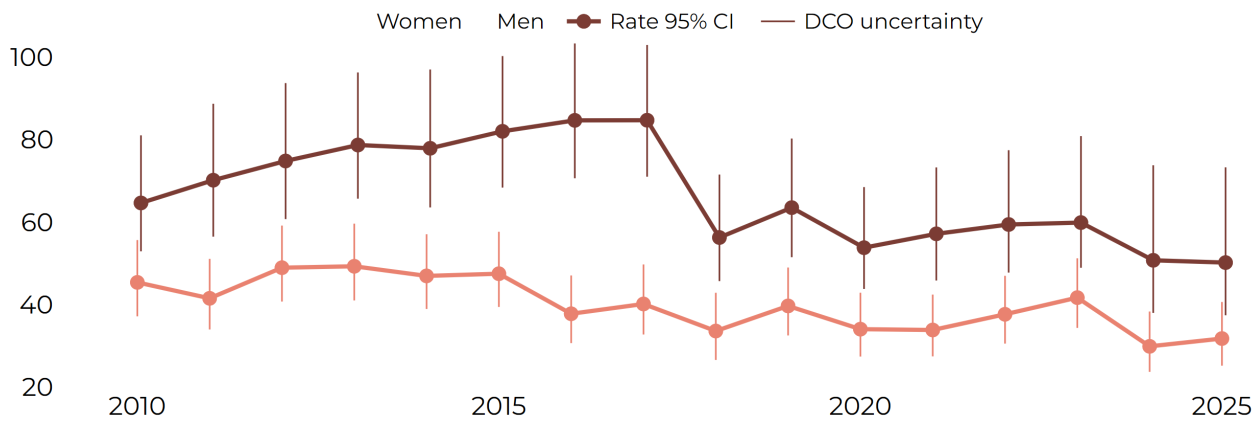
WHAT THIS MEANS

These rates allow the Heart mortality pattern to be compared over time without changes in population age structure driving the result. The Primary and Inclusive lines answer different definition questions; they are not confidence limits around one estimate.

Heart deaths: women and men

Primary age-standardised mortality rates per 100,000.

All sexes	Women	Men
39.3	31.7	50.2



Latest five complete years

Primary ASMR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	43.4	47.8	51.0	37.8	39.3
All sexes 95% CI	37.5 - 50.3	41.4 - 55.3	44.4 - 58.7	32.2 - 44.4	33.5 - 46.2
Women estimate	33.8	37.6	41.7	29.9	31.7
Women 95% CI	27.4 - 42.4	30.5 - 47.0	34.3 - 51.2	23.7 - 38.3	25.2 - 40.6
Men estimate	57.2	59.4	59.9	50.7	50.2
Men 95% CI	45.8 - 73.3	47.7 - 77.4	48.9 - 80.9	38.0 - 73.8	37.4 - 73.3

WHAT THIS MEANS

The women-and-men comparison uses the Primary definition and age-standardised rates, so it is more informative for service planning than raw counts alone. The chart identifies patterns worth following up, not causes.

Stroke deaths

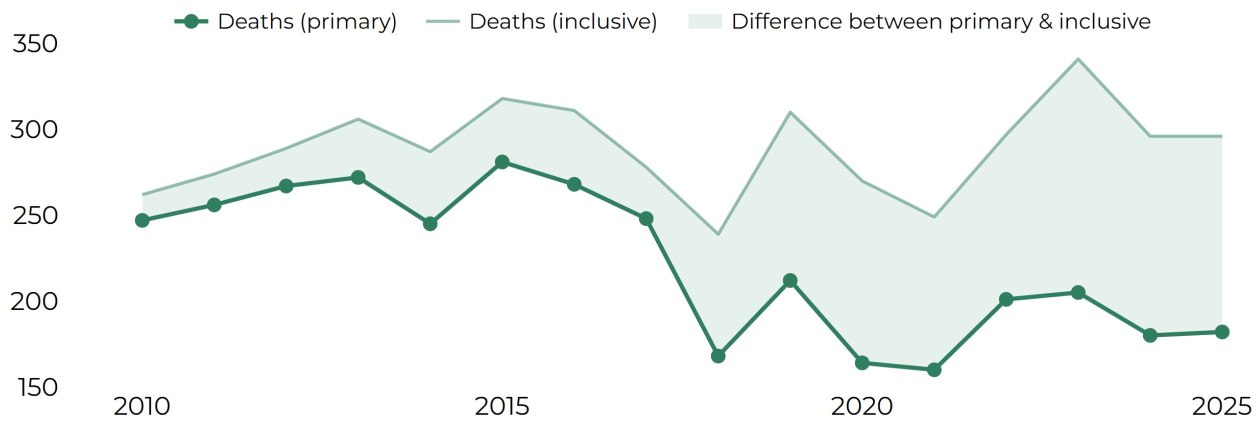
Primary = Clear + Likely CVD deaths. Inclusive = Clear + Likely + Possible CVD deaths.

Primary deaths

182

Inclusive deaths

296



Latest five complete years

Deaths	2021	2022	2023	2024	2025
Primary	160	201	205	180	182
Inclusive	249	297	341	296	296

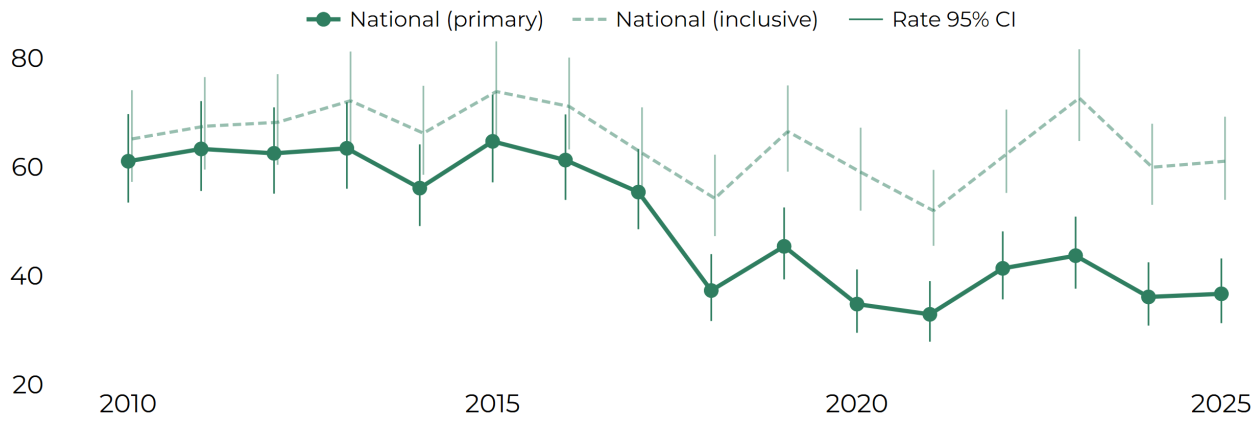
WHAT THIS MEANS

Stroke deaths are shown using both the Primary and Inclusive definitions. This gives decision-makers a transparent view of how the reported total changes when Possible CVD deaths are included.

Stroke mortality rates

Primary and Inclusive age-standardised mortality rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Primary ASMR	Inclusive ASMR
36.7	61.1



Latest five complete years

ASMR per 100,000	2021	2022	2023	2024	2025
Primary estimate	32.9	41.4	43.7	36.1	36.7
Primary 95% CI	27.9 - 39.0	35.7 - 48.2	37.6 - 50.9	30.8 - 42.5	31.3 - 43.2
Inclusive estimate	52.0	62.4	72.7	60.0	61.1
Inclusive 95% CI	45.5 - 59.5	55.2 - 70.6	64.8 - 81.7	53.1 - 68.0	54.0 - 69.3

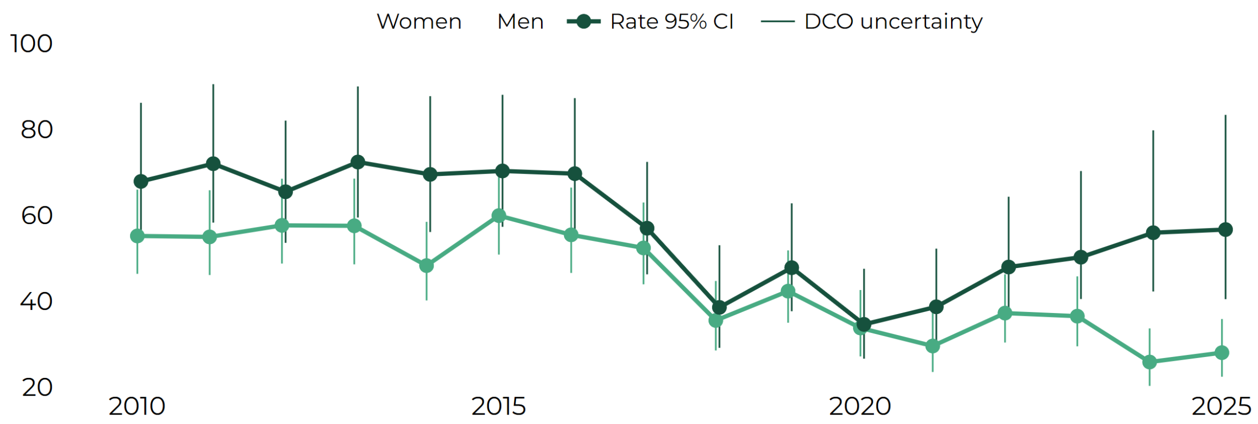
WHAT THIS MEANS

These rates allow the Stroke mortality pattern to be compared over time without changes in population age structure driving the result. The Primary and Inclusive lines answer different definition questions; they are not confidence limits around one estimate.

Stroke deaths: women and men

Primary age-standardised mortality rates per 100,000.

All sexes	Women	Men
36.7	28.0	56.7



Latest five complete years

Primary ASMR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	32.9	41.4	43.7	36.1	36.7
All sexes 95% CI	27.9 - 39.0	35.7 - 48.2	37.6 - 50.9	30.8 - 42.5	31.3 - 43.2
Women estimate	29.5	37.2	36.5	25.8	28.0
Women 95% CI	23.5 - 37.8	30.3 - 46.2	29.5 - 45.7	20.2 - 33.6	22.4 - 35.8
Men estimate	38.7	47.9	50.2	55.9	56.7
Men 95% CI	30.0 - 52.2	38.1 - 64.3	40.5 - 70.3	42.2 - 79.8	40.4 - 83.4

WHAT THIS MEANS

The women-and-men comparison uses the Primary definition and age-standardised rates, so it is more informative for service planning than raw counts alone. The chart identifies patterns worth following up, not causes.

Sources: BNR public CVD-event release cvd_2026_01; BNR public mortality release mort_2026_07.

3 | How complete is the picture?

Events identified through death records

Each tile gives the estimated additional DCO contribution as a percentage of the Primary national event estimate. The latest 15 complete years are shown. Green is lower reliance, amber moderate and red higher; the displayed number remains the main information.

	2011		2013		2015		2017		2019		2021		2023		2025
All CVD	19	17	10	15	19	17	13	12	12	12	13	14	13	16	8
Heart	14	7	11	15	20	12	15	12	14	12	15	17	18	19	8
Stroke	20	21	9	15	19	19	11	12	10	12	12	13	11	14	9

WHAT THIS MEANS

In 2025, the estimated additional DCO contribution was about 8.5% of the Primary national event estimate for All CVD, Heart and Stroke. The similarity across the three groups is useful context: death-record ascertainment contributes meaningfully to the national estimate, but it is not the dominant component of the 2025 event total.

Reliance on Possible deaths

Possible-only deaths as a percentage of the Inclusive annual mortality count. Green is lower reliance, amber moderate and red higher; the percentage remains the main information.

	2011		2013		2015		2017		2019		2021		2023		2025
All CVD	12	13	14	17	15	16	13	29	30	33	31	30	32	36	37
Heart	18	18	16	20	18	19	16	28	28	28	26	29	24	32	36
Stroke	7	8	11	15	12	14	11	30	32	39	36	32	40	39	39

WHAT THIS MEANS

Possible-only deaths accounted for 37.4% of the 2025 Inclusive All-CVD total. The corresponding proportions were also substantial for Heart and Stroke, at about 36% and 39%. These are sensitivity indicators for cause-of-death classification, not scores of whether the mortality data are good or bad.

4 | Methods

What this report measures

The standard section uses only the declared approved public CVD-event and mortality releases. Published rates, confidence intervals, linkage bounds and rolling comparators are read from those releases. Simple percentages used to explain DCO or Possible-death reliance are presentation summaries of published aggregate counts; no confidential data are reopened and no surveillance rate is recalculated. Asterisks denote values protected by the published disclosure-control rules.

Population, period and unit of analysis

This report describes eligible cardiovascular events and deaths among Barbados residents. Event statistics count events rather than necessarily unique people: one person may contribute more than one eligible event. Results use complete annual periods from the two declared public releases. Monthly information is used only for the clearly labelled contextual chart in the Year in Brief.

Measures used in Chapters 1 to 3

Report measure	What is counted or compared	Principal source	Denominator
Event counts	Eligible CVD events	Hospital records, with annual DCO-enhanced estimates	None
Event rates	Eligible events relative to population	Event release and approved population data	Matching resident population
Event patterns	Events by sex or broad age group	Event release	Relevant eligible event group
Death counts	Deaths meeting a BNR mortality definition	Death-certificate information	None
Mortality rates	Classified deaths relative to population	Mortality release and approved population data	Matching resident population
Mortality patterns	Classified deaths by sex or broad age group	Mortality release	Relevant classified-death group
Quality measures	Reliance on additional or less-certain information	Approved release aggregates	Measure-specific eligible total

The annual report is a reader of approved public statistics, not a second analytical workflow. It does not reopen confidential records, redo linkage, recalculate rates or confidence intervals, or reconstruct protected values.

How CVD events are identified

Consistent event rules allow changes over time to be interpreted more fairly. Records enter the event series only after the approved disease, residence, date, source and duplicate-resolution rules have been applied.

Heart, Stroke and All CVD

Heart and Stroke are the two broad event families used in this report. All CVD combines eligible events from those families according to the approved classification hierarchy, so the same event is not counted twice merely because more than one cardiovascular term is present.

Heart and Stroke event definitions

In the event series, Heart is the BNR reporting label for eligible acute myocardial infarction (heart-attack) events. Stroke is the BNR reporting label for eligible acute stroke events. These are clinical event definitions, not simply any record that contains a heart- or stroke-related word. The maintained confidential workflow applies the approved diagnostic, residence, date and duplicate-resolution rules before assigning an event to either family.

Reporting family	What it represents	What it does not mean
Heart	Eligible acute myocardial infarction events under the maintained BNR clinical rules	Every cardiovascular presentation, cardiac symptom or certificate mention
Stroke	Eligible acute stroke events under the maintained BNR clinical rules	Every neurological presentation, vascular risk factor or certificate mention

Decision	How it affects the published statistic
Disease definition	The record must meet the approved Heart or Stroke criteria.
Residence	The event must relate to a Barbados resident under the stated rule.
Event date	The approved event date places the event in its reporting period.
Duplicate resolution	Records describing the same event episode are resolved before counting.
Repeat event	A later eligible event may be counted separately when it falls outside the approved 28-day event window.
Analytical completeness	A record must contain the information required for the selected measure.

Events and people

The unit is an event, not necessarily a unique person. Records within an event window are assessed as part of one episode; a later event can contribute another count when it independently meets the rules. Event counts should therefore not be described as numbers of different people.

Hospital-recorded ascertainment

Hospital-recorded statistics use eligible events identified from hospital information. The Queen Elizabeth Hospital is Barbados's only tertiary hospital, making this series an important view of serious recognised CVD events. It is not assumed to represent complete national ascertainment: access to care, referral, diagnosis and recording can all affect what is observed.

WHEN READING THE REPORT | Compare event series only when their disease definition, source coverage, period and population are compatible.

Extending the picture using death records

Some eligible cardiovascular events are identified only through death information. These death-certificate-only events broaden the annual picture beyond hospital records, but they introduce additional questions about classification and whether a death record matches an event already counted by the hospital.

What is a DCO event?

A death-certificate-only (DCO) event is an eligible event identified from death information without a matched eligible hospital-recorded event. DCO-enhanced event counts and rates are annual only. Monthly and quarterly event views remain hospital-recorded because death-record ascertainment closes after the year ends.

How linkage is undertaken

Stage	Plain-language description
1 Prepare	Eligible hospital events and the Inclusive eligible death pool are prepared in the confidential environment.
2 Match	A short sequence of deterministic passes uses approved combinations of identifiers, dates and event windows, beginning with the strongest unique matches.
3 Resolve	Convincing matches are accepted. Ambiguous or weaker candidates remain unresolved rather than being forced into a match.
4 Aggregate	Primary and Inclusive aggregate event representations are derived from the mortality evidence class; identifiers never enter the public release.

Published national representations

Published series	Source coverage	Interpretation
Hospital-recorded	Eligible hospital events	Observed serious events recorded through the hospital source
Primary national	Hospital events plus DCO contribution based on Clear + Likely mortality evidence	Main broader annual estimate
Inclusive national	Hospital events plus DCO contribution that also includes Possible mortality evidence	Broader classification-sensitivity estimate

Linkage range

Where unresolved linkage could alter a national estimate, the public release carries approved lower, central and upper aggregate values. The lower value is the more conservative representation, the central value is the published reporting estimate and the upper value reflects the broader plausible contribution allowed by the linkage method. This range describes linkage uncertainty, not ordinary statistical variation.

WHEN READING THE REPORT | A wider linkage band means that unresolved record linkage has more influence on the national estimate. It is not a 95% confidence interval.

How CVD deaths are classified

BNR mortality surveillance uses death-certificate information to create a consistent practical cardiovascular classification when a formally assigned national underlying cause of death is not available to the Registry for routine surveillance. A cardiovascular term appearing anywhere on a certificate does not automatically mean that CVD is treated as the underlying cause.

How the certificate is interpreted

The method considers the recorded wording, whether information appears in Part I or Part II, the order of conditions in Part I, a limited set of documented mortality-coding principles and the strength of the available cardiovascular evidence. Part I is used to record the causal sequence leading directly to death; Part II records other important contributing conditions. Line position therefore provides context and is not treated as a simple ranking of diagnoses.

Evidence classes

Class	BNR interpretation
Clear	Strong evidence that the death belongs in the relevant BNR cardiovascular mortality group.
Likely	Good evidence, although some uncertainty remains.
Possible	Cardiovascular attribution is plausible, but uncertainty is material.
Mention only	Relevant wording is present but does not support approximate underlying-cause attribution.
No evidence	No qualifying cardiovascular evidence is identified by the approved rules.

Heart, Stroke and All CVD

Heart includes qualifying ischaemic and coronary heart-disease evidence and is not limited to certificates that literally say heart attack. Stroke includes qualifying acute ischaemic stroke and non-traumatic cerebral haemorrhage patterns under the approved rules. All CVD combines the resolved Heart and Stroke classifications according to the maintained hierarchy.

Primary and Inclusive definitions

Definition	Evidence included	How to interpret it
Primary	Clear + Likely	The main BNR mortality reporting definition
Inclusive	Clear + Likely + Possible	A broader sensitivity definition that includes more uncertain cardiovascular attribution

Inclusive contains Primary. The two definitions are alternative views of the same mortality burden and must not be added together. Their difference shows sensitivity to the inclusion of Possible deaths; it is not a statistical confidence interval.

Relation to official cause-of-death statistics

The BNR method is a structured approximate-underlying-cause classification for surveillance. It does not implement the full international process for selecting and coding an official ICD-10 underlying cause of death. BNR results should therefore not be assumed to be directly equivalent to official national or international mortality series, even when their labels appear similar.

WHEN READING THE REPORT | Primary is the main series. Inclusive shows how the result changes when Possible cardiovascular deaths are included.

How the report measures CVD

Counts and percentages

A count is the number of eligible events or classified deaths. A percentage describes how a defined eligible total is divided across groups. The denominator must always be checked: a percentage among registered events or classified deaths is not a person's probability of experiencing or dying from CVD.

$$C = \sum_i I_i$$

Here C is the count and I_i equals one when record i is an eligible event or death and zero otherwise.

Percentages

$$p = \frac{n_g}{N} \times 100$$

Here p is the percentage, n_g is the selected group and N is the relevant eligible total.

Crude population rate

$$R = \frac{E}{P} \times 100{,}000$$

Here R is the rate, E is the eligible event or death count and P is the matching resident population. A crude rate relates the observed number to the matching population and period. It is useful for describing actual population experience, but comparisons can be affected by differences in age structure.

Direct age standardisation

$$ASR = \sum_a w_a r_a$$

Here ASR is the age-standardised rate, r_a is the rate in age group a and w_a is that group's standard-population weight. The report uses the WHO World Standard Population 2000-2025. Standardisation improves comparisons by applying the same age distribution to every group or year; it does not estimate an event count or remove uncertainty from ascertainment, classification or small numbers.

Denominators and reported forms

Measure	Numerator	Denominator	Main reported forms
Event rate	Eligible hospital or DCO-enhanced events	Matching Barbados resident population	Crude and age-standardised
Mortality rate	Deaths under the selected BNR definition	Matching Barbados resident population	Crude and age-standardised
Women and men	Sex-specific events or deaths	Resident population of the same sex	Separate published rates
Age pattern	Events or deaths in a broad age group	Relevant eligible total or age-specific population, as labelled	Count, percentage or crude age-specific rate

Population denominators use the approved UN World Population Prospects 2024 Barbados series carried by the public workflow. Every numerator is paired with the denominator for the same year and, where relevant, the same sex and age group.

WHEN READING THE REPORT | Counts describe volume. Rates support population comparison. Age-standardised rates support fairer comparison across differently aged populations, but are not actual counts.

Understanding uncertainty, comparisons and time

No single interval or comparator describes every source of uncertainty. This report keeps statistical variation, unresolved record linkage and mortality-classification sensitivity separate so that each can be interpreted for the question it answers.

What is shown	How it may appear	What it means
Statistical 95% confidence interval	Whiskers or a transparent band, according to the graphic	Statistical variation around the published rate under its stated method
DCO linkage range	Pale transparent band	How unresolved deterministic linkage may affect an annual national event estimate
Primary and Inclusive mortality	Separate lines or values	Sensitivity to including Possible cardiovascular deaths
Previous-five-year mean	Comparator line or value	Historical context, not an uncertainty interval or forecast

Statistical confidence intervals

Published crude-rate intervals use the exact Poisson (Garwood) method where specified. Published directly age-standardised mortality intervals use the Fay-Feuer gamma method. Event-rate rows carry the interval method supplied by the approved event release. The report displays the released limits and does not recalculate them.

What these intervals do not capture

A statistical interval does not capture every effect of source coverage, diagnosis, health-care access, late records, mortality classification, unresolved linkage or population estimates. These considerations remain important even when a confidence interval is narrow.

Comparators and complete periods

Annual count comparisons use the published mean for the same stratum over the previous five calendar years where it is available. Monthly historical lines provide seasonal context. Neither is a confidence interval, target or prediction. The standard annual results use complete calendar years; partial monthly or quarterly rows remain appropriate for operational dashboards but are not treated as complete annual findings here.

Data availability

Series	First complete annual year	Latest complete annual year	Release used
CVD events	2010	2025	cvd_2026_01
CVD mortality	2010	2025	mort_2026_07

Release date and latest observation period are different. An approved release published later can legitimately contain data ending in an earlier complete year. Material revisions should replace the superseded current value through the controlled release process and remain traceable through the release record.

WHEN READING THE REPORT | Do not combine a statistical confidence interval, a DCO linkage range and the Primary-Inclusive difference into one overall interval. They answer different questions.

Data quality, confidentiality and publication

What the Chapter 3 measures describe

The Chapter 3 displays describe how strongly national event estimates rely on death-record ascertainment and how strongly mortality totals rely on the less-certain Possible class. They are quality and sensitivity summaries, not measures of individual risk and not judgements that a particular record is correct or incorrect.

DCO contribution	estimated additional DCO events / Primary national event estimate × 100
Possible-death reliance	Possible-only deaths / Inclusive mortality count × 100

Completeness

Where field completeness is reported, the numerator is the number of eligible records containing the required information and the denominator is the number for which that information should be available. A field that is genuinely not applicable is excluded from that denominator rather than counted as missing. Changes in completeness can alter apparent patterns without a corresponding change in disease burden.

Disclosure control

BNR publishes aggregates rather than patient-level records. Under bnr_sdc_v1, exact supporting counts from 1 to 5 require primary suppression; a true zero is not automatically suppressed solely because it is zero. Secondary suppression protects values recoverable from totals, while derived and temporal protections address calculated outputs and differencing across releases. The complete proposed release is reviewed by a person before approval. Suppressed means not published, not zero, and the annual report never reconstructs a protected result.

Two controlled production pathways

Layer	Controlled stages	Boundary
Metric releases	Prepare → calculate → review → approve → publish	Creates approved aggregate CVD-event and mortality products from controlled analytical inputs
Annual report	Build candidate → approve candidate → publish	Reads declared public releases; it does not repeat classification, linkage, calculation or suppression

Responsible interpretation

Changes may reflect health, health-care access, diagnosis, recording, completeness, population estimates or several factors together. Compare like with like; distinguish events from people and hospital-recorded results from national estimates; consider all three forms of uncertainty separately; and interpret small annual changes cautiously. Surveillance patterns can guide decisions and questions, but do not by themselves prove why a change occurred.

WHEN READING THE REPORT | Definitions, source coverage, period, denominator, completeness and uncertainty are part of every result, not optional footnotes to the number.

Special chapter | 2025

The BNR Refit

From audit to controlled reporting

Why the refit was needed

The BNR process audit began as a review of the post-REDCap Stata and reporting workflow. It expanded when upstream data and governance problems were found to affect downstream analysis. The audit highlighted four recurring risks: no single definitive cumulative dataset, persistent missingness in important fields, legacy Stata code that mixed cleaning with analysis, and no consistent dataset release and sign-off process.

Audit finding	Why it mattered
No definitive cumulative dataset	Parallel copies made it difficult to know which dataset was authoritative.
Persistent missingness	Missing identifiers and event dates weakened linkage and downstream analysis.
Legacy Stata code	Cleaning and analysis were intertwined, increasing the need for manual editing between exports.
No consistent release sign-off	Unverified data could enter analysis or reporting without a formal release decision.

A different operating model

The refit response is deliberately architectural rather than a collection of one-off fixes. Data releases are declared and versioned; Stata produces reproducible metric outputs; automated checks and disclosure control happen before human approval; and publication is separated from calculation. The result is a controlled path from source data to public reporting without removing the analyst from the decision process.

1 DATA FREEZE	A named, versioned source release defines the analytical cycle.
2 COMPUTE	Readable Stata programs create the approved surveillance measures and structured outputs.
3 REVIEW	Automated QA and disclosure checks prepare a fixed candidate for human inspection.
4 APPROVE	An authorised BNR reviewer confirms analytical plausibility, disclosure safety and publication readiness.
5 PUBLISH	Only the approved payload is promoted to the public release and website mirror.

The human role is retained

Automation is used for repeatable calculation and validation, not to remove professional judgement. BNR staff still decide whether the declared source releases are appropriate, whether results are plausible, whether interpretation is suitable, and whether the candidate should be released. Generated datasets and reports are corrected by changing source data or version-controlled code and rerunning the workflow, not by editing public files by hand.

From legacy practice to controlled reporting

Area	Legacy risk identified by the audit	Refit direction
Dataset identity	Multiple parallel copies with uncertain authority	Declared release IDs, versioned files and controlled promotion
Analytics	Cleaning and analysis combined in evolving scripts	Separated preparation, metric calculation and publication layers
Quality control	Problems often discovered late in analysis	Automated QA plus visible human review before approval
Disclosure	No single repeatable publication gate	Deterministic disclosure control bound to the reviewed candidate
Reporting	Manually assembled outputs with limited reproducibility	Approved public datasets feed dashboards, updates and this annual report

What this changes for annual surveillance

This annual report is itself part of the refit. Its standard surveillance section does not reopen confidential source data or recalculate measures that already passed through the event and mortality workflows. It reads the declared approved public releases, combines them into a coherent annual narrative, and then enters its own review and publication gate. The Special chapter remains deliberately flexible so that each annual report can examine one topic in greater depth without weakening the repeatable surveillance core.

The practical objective is modest but important: the same published statistic should mean the same thing whether it is encountered in a dataset, dashboard, rolling update or annual report.

Sources

BNR Process Audit: Findings (Information Hub); BNR Surveillance Automation and Integrated Digital Reporting, Terms of Reference v1.0 (26 February 2026); current BNR Operations and Technical documentation.

About this report

Report identifier: bnr_cvd_annual_report_2025_v19

Report version: v19

CVD-event release: cvd_2026_01

Mortality release: mort_2026_07

The PDF is built from the declared approved public releases. Publication remains subject to the annual report Step 2 review and Step 3 publication controls.